

Business Math I (MT1002)

Final Exam

Date: December 26th 2025

Course Instructor(s)

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Total Time : 3 Hour

Total Marks: 75

Total Questions: 6

25L-5602

Roll No

B5F+1C

Section

Momina

Student Signature

CLO # 2: Apply fundamental concept of Transcendental functions, Exponential functions, Logarithmic functions.

CLO # 5: Solutions for Optimization problems

Q.No.1:

A small manufacturing firm produces customized notebooks for corporate clients. The firm incurs a fixed daily cost of Rs12000 for rent, utilities, and administration. In addition, the variable cost of producing each notebook is Rs 180. If Q is the number of notebooks produced & sold per day, the total revenue (in Rs) can be modelled by a quadratic function. The revenue is Zero when zero notebooks are sold ($R=0$ at $Q=0$), and reaches a maximum when 250 notebooks are sold per day.

- Write down the Linear cost function $C(Q)$. (2)
- Model a suitable quadratic revenue function $R(Q)$, consistent with the given information. (3)
- Hence find the profit function $\pi(Q)$. (2)
- Find Marginal cost & marginal Revenue functions. State whether the firm is a monopolist or a perfectly competitive firm. (3)
- Using marginal functions, determine the level of output at which profit is maximized. Find maximum profit. (3+1)
- Determine the intervals on which profit is increasing/decreasing. (4)
- Find the break-even points. (2)
- Verify your working by sketching Revenue & Cost function on same diagram. Label maximum profit, inc/dec profit, break-even pts on the diagram. (5)

CLO # 5: Solutions for Optimization problems

Q.No.2:

- Given the constraints Sketch the feasible region & find the corner points.
 $2x + 3y \leq 60$; $4x + y \leq 40$; $x \leq 12$; $y \leq 18$; $x \geq 0$; $y \geq 0$. (6)
- A furniture manufacturer produces tables & chairs. Each table requires 4 hours of carpentry & 2 hours of finishing, while each chair requires 2 hours of carpentry & 3 hours of finishing. The firm has at most 80 hrs of carpentry time and at most 60 hrs of finishing time available per week. The profit on each table is Rs 3000, and the profit on each chair is Rs 2000. Determine the number of tables and chairs produced & sold per week, for which the total Profit is maximized. Formulate the objective function & constraints of the problem only. (4)

CLO # 6: Gaussian Elimination method for solving matrices

Q.No.3:

Solve the following systems of equations by the Gaussian elimination method (10)

$$\begin{aligned} 10P_1 - 5P_2 + 3P_3 &= 55 \\ -7P_1 + 14P_2 - 5P_3 &= 27 \\ -P_1 - 5P_2 + 7P_3 &= 18 \end{aligned}$$

CLO # 3: Financial Mathematics Problems

Q.No.4:

(a) A company manufactures deckchairs for a 10-week period each year. Production starts with 600 chairs in week 1 and increases by 5% for each subsequent week. Use series to calculate:

- (i) The number of chairs manufactured in week 7.
(ii) The total number of chairs manufactured during the 10 weeks. (5)

(b) A bank pays 7.5% interest, compounded quarterly. How long will it take for £10 000 to grow to £20 000? (5)

CLO # 5: To discuss the techniques of Differentiation and Integration and its applications for business students

Q.No.5:

(a) Differentiate the following: (5)

(i) $y = xe^{\ln(3x)}$ (ii) $y = (2x + 1)^2$

(b) Integrate the following: (5)

(i) $\int (\frac{1}{e^{2t}} + 2e^{2+t^2} - 9)dt$ (ii) $\int \frac{15}{(2Q-7)^4} dQ$

CLO # 2: Apply fundamental concept of Transcendental functions, Exponential functions, Logarithmic functions.

Q.No.6:

(a) Solve the following equation: $\log Q - \log \frac{Q}{Q+1} = 10$ (5)

(b) 154 students attend a maths lecture. If 22 students are absent, calculate the percentage of students absent.

(c) The 2016 price of a basic computer will be 35% lower than the 2013 price. If the 2016 price is £910, calculate the price in 2013. (2+3)

2016 65% 910
2013 100% X